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Investigating the Jettability of Toluene in a Piezo Inkjet

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Investigating the Jettability of Toluene in a Piezo Inkjet

by

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Report

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Abstract

Investigating the Jettability of Toluene in a Piezo Inkjet

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This report starts by introducing the working principles of a single nozzle piezo inkjet system. The current voltage waveform calibration process for a piezo inkjet is time consuming and subsequently a cause for material waste. Recent research has demonstrated that the use of real-time imaging of in-flight drops integrated with genetic algorithms (GAs) can lead to improved waveforms that provide enhanced system performance. In this work, the same imaging system is used to reproduce known results for simple fluids such as water, and manual waveform tuning was used instead. This framework was then extended to nonpolar solvents.

The experiments presented in this project investigated the jettability of toluene. Toluene is a nonpolar liquid that has raised interest about its viability as an ink used in piezo inkjets. The manual waveform tuning process was centered around adjusting key parameters such as dwell time and dwell voltage. As these parameters were tweaked, real-time imaging was used to observe the drop formation at the orifice of the inkjet

nozzle. Once the key parameters were optimized, a droplet of toluene was generated. Unipolar, bipolar, and tripolar actuation waveforms were investigated to determine if a droplet of toluene was able to be dispensed. Bipolar and tripolar waveforms produced droplets that were dispensed at irregular frequencies along with physical deformities. This work found that using a unipolar actuation waveform produced the best results, generating consistent spherical toluene droplets with an average drop volume of 30.2 pL. The optimal waveform parameters were as follows: a rise time of 11 μ s, a dwell time of 36 μ s, a fall time of 9 μ s, and a dwell voltage of 60 V.

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Chapter 1: Introduction

Inkjet printing is a manufacturing method for micro-/nanofabrication by using ink droplets to form patterns onto a substrate. This method is revered for its ability to control droplet volume with picolitre accuracy and for precisely placing a droplet onto a substrate with micrometer accuracy [1]. Common applications include coding/marketing of consumer products and patterning for electronic devices such as displays. But unique applications enabled by inkjet printing include fabrication of biosensors, microdeposition of proteins on cellulose, flexible electronics, and printed organic light-emitting diodes (OLEDs) for mass market use [2,3]. These complex applications are achieved using experimental inks, such as biomaterials and nonpolar fluids. Nonpolar fluids are made up of molecules that do not have positive or negative charges. They have a challenging rheology, exhibiting low surface tensions and evaporate quickly, but their jettability in piezo inkjets is raising interest. Piezo inkjet systems using nonpolar fluids as a dispensing ink is an emerging method for patented applications of practical interest, such as producing opaque images on a substrate and developments in spray ink printing [4,5].

There are many types of inkjet technology, this report focuses on a piezo inkjet system. A piezo inkjet is a drop-on-demand inkjet (DOD) with an annular piezoelectric transducer (PZT) that is driven by a voltage pulse. The drive voltage can come in various waveform shapes, which will affect the physical characteristics of the droplet being ejected from the nozzle. Depending on the nozzle dimensions, ink rheology, and the desired drop volume and drop velocity, a specific waveform must be generated to fit these parameters.

Waveform tuning is a process that seeks to generate the optimal input waveform to satisfy the desired droplet properties, which are typically drop volume and drop velocity. This calibration process can be achieved through a variety of computing techniques. One approach

uses a genetic algorithm that cycles through iterations of waveforms to mutate towards a final solution, but this can take hours to complete and wastes a vast amount of fluid. Improving the initial starting point for the waveform optimization would help reduce calibration time and material waste. A tuning methodology that can decrease material waste is highly desirable as different types of experimental inks are starting to be introduced to piezo inkjet systems.

Nonpolar fluids refer to liquids that are comprised of molecules with no net electrical charge. In nonpolar covalent bonds, electrons are distributed evenly. Common examples are gasoline, oils, and solvents like toluene and xylene. Nonpolar liquids tend to be difficult to inkjet because solvent ink can evaporate quickly and clog the nozzle, and can cause electrode corrosion in piezo heads [6,7]. Due to these challenges, this project aims to explore the jettability of nonpolar solvents in piezo inkjets. To test its jettability, the parameters that make up the actuation waveforms are manually adjusted and the behavior at the nozzle orifice is observed until a proper droplet is ejected. Manual tuning gives a better understanding on how the variation of each individual waveform parameter impacts the process of the droplet formation. The solvent will be tested to determine if a droplet is ejected out of the orifice of a piezo inkjet nozzle.

This project investigated the jettability of nonpolar solvents in piezo inkjets by determining whether droplets could be generated with toluene. Experimental results using manual waveform tuning promise to give more insight moving forward with automated waveform tuning in future projects. Further, the results help provide more data for reducing an inkjet calibration time when using experimental inks.

Chapter 2: Background Information

2.1 Inkjet Technology Overview

Inkjet technology is classified into two main categories, continuous inkjet (CIJ) and Drop-On-Demand inkjet (DOD). A CIJ has a continuous stream of ink droplets ejected out of a nozzle, typically ejected out of an ink column sustained under constant pressure. In this system there are excess droplets that might not land directly on the substrate, they are caught in a gutter and recycled. CIJ printing is not a common practice for printed electronics because the recycling process exposes the ink to the environment and may result in ink contamination. DOD inkjets eject ink droplets on demand with a unique type of actuation mechanism. The most common types of actuators for DOD inkjets are electrostatic, thermal, and piezo inkjets [8].

Inkjets using electrostatic actuation consist of a ground electrode located inside the nozzle and an extraction electrode plate located between the substrate and the outside of the nozzle. When a high voltage pulse is sent through the outer electrode, the electrostatic force between it and the ground causes the liquid to form a microdroplet. When the force is stronger than the surface tension at the tip of the nozzle, the ink breaks up into microdroplets and is ejected onto a substrate. The main advantage for this type of actuation is its ability to make droplets smaller than the size of the nozzle, an area at which thermal and piezo inkjets lack [9]. This technique is still developing since it is not as mature as piezo or thermal inkjets, but the investment and running cost is much higher than a piezo inkjet.

A thermal inkjet (TIJ) uses a heating element to flash heat ink to form a vapor bubble that ejects an ink droplet. This is achieved by first applying heat to a thin film resistor to heat up the ink solution. A vapor explosion in the fluid nucleates a vapor bubble that rapidly expands. This

expanding bubble pushes the ink out of the nozzle to eject a droplet out of its orifice. After the droplet is dispensed from the nozzle, the original vapor bubble collapses and the chamber fills with additional ink again to repeat the process [10]. The TIJ method has the advantages of low manufacturing cost and print quality when applied to outputting photos and coding for consumer products. TIJ printer heads can be unreliable because of heater breakdowns due to heat stress, residual bubbles inside the nozzle, and jetting failure due to ink viscosity increase [11].

A piezo inkjet system consists of a piezoelectric transducer (PZT) that is actuated by a voltage pulse. The frequency of the voltage pulse can be tuned, usually between 1kHz-20kHz, and can be applied as different waveform shapes such as unipolar, bipolar, and tripolar. The actuation waveform is applied to the PZT through electrodes connected to its inner and outer surface, and piezo actuation creates pressure waves within the ink channel that is inside the nozzle. These pressure waves created are responsible for driving the ink out of the nozzle, and as the ejected fluid leaves the nozzle orifice it is pulled in a spherical drop by surface tension forces [12,13]. The main drawback is that piezo print heads are more expensive and have fewer print heads per printer compared to a TIJ printer. But a piezo inkjet's merit comes from its ability to have variable and precise droplet sizes for a wide range of applicable inks. The range of ink solvents that can be used in a piezo inkjet are much broader than thermal and electrostatic inkjets, where unconventional inks can vary from biomaterial fluids to nonpolar solvents.

2.2 Piezoelectric Transducers

Lead zirconate titanate (PZT) ceramics exhibit the unique property of the inverse piezoelectric effect. When an electric field is applied to these kinds of ceramics, physical deformation occurs [14]. Piezo inkjets take advantage of piezoelectric ceramics with different

deformations modes to actuate the motion required to eject fluid out of a nozzle. These unique deformations contribute to four main types of piezo actuations: squeeze, bend, push, and shear.

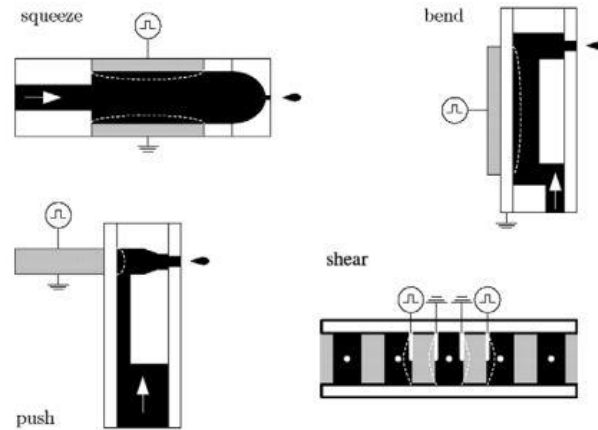


Figure 1. Piezo inkjet printhead technologies by deformation mode [8]

For the squeeze mode actuator, a PZT ceramic tubing is polarized radially across electrodes on the inner and outer surface of the ring-like structure. The deformation expands and compresses the nozzle to expel the droplet. Bend and push modes have electrodes parallel to the polarization of the piezomaterial, where bend modes cause a diaphragm to flex inwardly on a pressure chamber to squeeze the droplet out. Push mode has a rod pushing against a very thin diaphragm to eject the droplet. In a shear mode actuator, electrodes generate fields perpendicular to the polarization direction in the channel walls, which causes the upper halves to displace and force the lower halves to follow their motion. These displacements increase or decrease the channel volume to dispense a droplet [8].

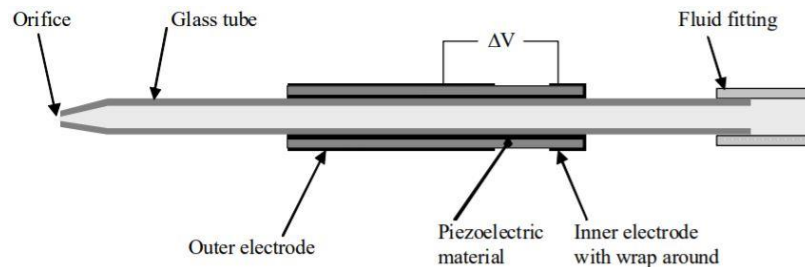


Figure 2. Schematic of MicroFab's Piezo Inkjet [13]

In this project, the experimental setup for the single nozzle inkjet uses the MicroFab dispenser shown in Figure 2. The dispenser configuration is MJ-AT-01-050, a micro jet (MJ) with a threaded fitting (AT) having an orifice diameter of 50 μm . The annular actuator has electrodes on its inner and outer surfaces. Small gauge wires are soldered to the outer and inner electrodes and the wires are loosely twisted and placed together in a connector that attaches to a MicroFab JetDrive electronics box to receive actuation waveforms.

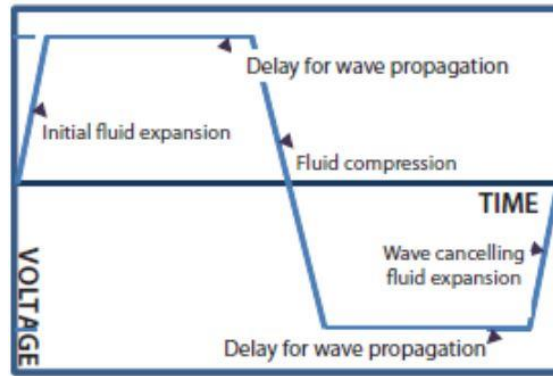


Figure 3. Bipolar waveform actuation shape with the ink's corresponding fluidic actions [12]

Figure 3 shows the process when a piezo element experiences a voltage pulse with a bipolar waveform, during the rise time the tubular PZT expands and creates a negative pressure in the nozzle chamber, which causes an initial fluid expansion. During the fall time the piezo element reduces the channel cross-section to its original size and continues compressing the fluid to amplify the positive pressure wave that drives the ink through the nozzle. Lastly, during the second rise time, the cross-section returns to its original state, and the process is repeated for every experienced waveform.

2.3 Parameterization of Actuation Waveforms

The actuation waveform sent to the piezo inkjet is a critical factor in the jetting process. A specific type of waveform will affect the inkjet's ability to dispense the droplet successfully

from the nozzle and will control the frequency at which they are ejected. Most importantly, the waveform will determine the quality of droplets so that droplet deformations can be avoided.

In order to achieve desired droplet characteristics, several trapezoidal driving waveforms have been developed, such as unipolar, bipolar, tripolar, and other complex waveforms [15,16]. The unipolar waveform is the simplest and most used waveform to generate a droplet.

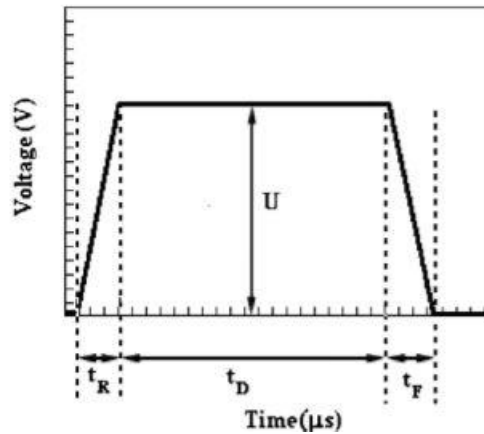


Figure 4. Profile of a unipolar waveform [16]

Based on Figure 4, the waveform parameters consist of a rise time (t_R) to reach a set voltage (U) that stays constant for a duration known as the dwell time (t_D) and is followed by a fall time (t_F). This figure has a symmetrical shape due to the rise and fall times being equal, but it is common to edit these durations to be unequal as well. The values for these parameters can vary, but typical rise and fall times can be as rapid as 1-10 μs while the dwell time could be about 20-30 μs . The amplitude, U , can range anywhere from 15-50 V.

A bipolar waveform, also known as a standard wave, consists of a positive trapezoidal pulse followed by a negative trapezoidal pulse. The first positive pulse is called a resonating pulse, it is capable of jetting drops but can generate residual oscillations that can result in droplet properties different from the previous drop. The negative pulse that follows is called a quenching pulse, which is used to dampen these residual oscillations [17].

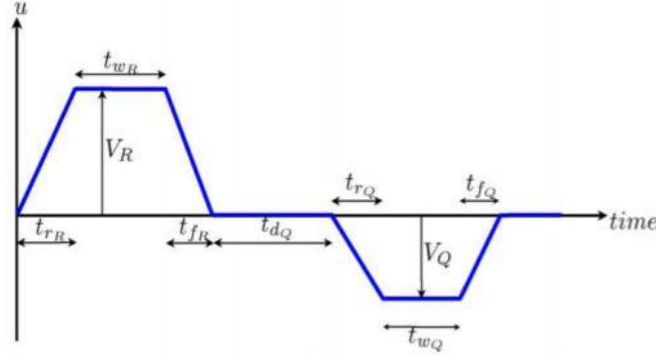


Figure 5. Profile of a bipolar waveform with a time interval, t_{dQ} , between the upper and lower pulse [17]

The parameterization for the bipolar waveform is distinguished by its resonating and quenching pulses. In Figure 5, the parameters that are optimized for both these waveforms are the rise time (t_r), the dwell time (t_w), the fall time (t_f), and the amplitude (V). The time interval between the resonating and quenching pulse is denoted as t_{dQ} . This parameterization is as detailed as possible as it accounts for each segment of the bipolar waveform shape, which allows for specific manipulation of these individual parameters.

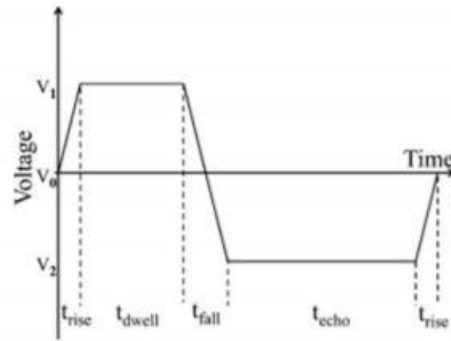


Figure 6. Profile of a bipolar waveform without a time interval, t_{dQ} [15]

In Figure 6, there is no time interval between resonant and quenching pulses, which is when $t_{dQ} = 0$, and the parameter between the pulses is simply known as t_{fall} . In Figure 6, the resonating pulse amplitude, V_1 , is known as the dwell voltage and the quenching pulse amplitude, V_2 , is known as the echo voltage [15]. Figure 6 is the standard waveform profile this

project focuses on, where the JetDrive software used in this project allows one to set both rise times, the fall time, and the dwell and echo voltage to any desired values.

Tripolar waveforms are slightly more complex as they introduce another trapezoidal pulse.

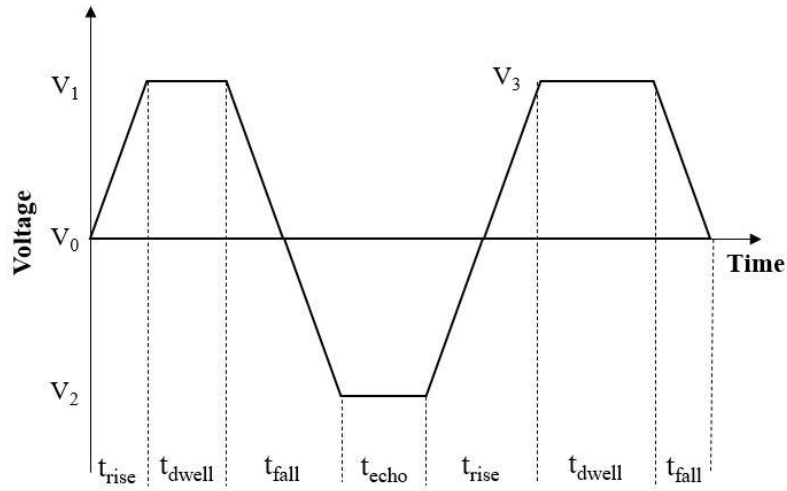


Figure 7. Profile of a tripolar waveform

In Figure 7, a few more parameters are added on top of what the bipolar waveform looked like, such as a second dwell time at V_3 that is followed by a second fall time. Generally, a higher-order waveform, such as this tripolar waveform, is used to induce complex pressure gradients at the orifice to generate smaller droplets [15].

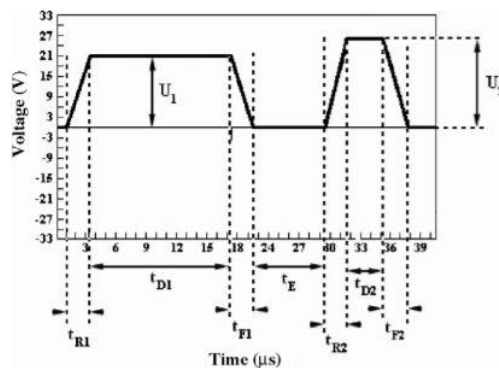


Figure 8. Profile of the M-shaped waveform, t_E represents interval time between the first and second pulses [16]

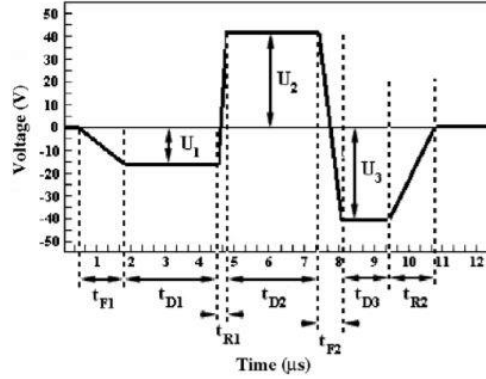


Figure 9. Profile of the W-shaped waveform [16]

In Figure 8 and Figure 9, these waveforms are called M-shaped and W-shaped waveforms respectively. Gan et al. developed these waveforms with the purpose of minimizing droplet deformities particularly for a PEDOT ink [16]. These two wave types illustrate the possibility of designing complex waveforms for specific applications. In this project these complex waveforms are not utilized, but it is worth noting that one can manipulate waveform parameters and its overall waveform shape at the user's desire.

2.4 Desired Droplet Characteristics and Potential Droplet Deformations

Due to the variety of applications a piezo inkjet can be used for, each different project will require different droplet properties. For example, the dimensions of the pattern being built on a substrate may require limited drop volumes. Or perhaps the substrate may be constrained to be a certain distance away from the nozzle orifice which may lead to a need for a particular drop velocity to ensure placement accuracy. Drop volume and drop velocity are droplet characteristics that a user is seeking when jetting various inks.

Jetted drops are radially symmetric when ejected from a circular nozzle orifice with uniform surface properties. Drop volumes can be estimated using disc integration, where in a droplet image a row of pixels is rotated about a radial axis [15]. Drop velocity can be estimated by tracking the position of the droplet over time, specifically drop centroid positions.

Desired droplet characteristics are optimized using a waveform tuning process, but there are a few issues to avoid along the way. The main type of droplet deformations to avoid are satellite drops. Ligament satellites, shown in Figure 10 on the left, form when the ligament behind a drop detaches from the primary drop below it and forms a satellite drop. The ligament has the possibility of splitting into multiple drops and the tension that splits the ligaments can also accelerate each drop towards each other resulting in a forward-merging of the drops. Meniscus satellites, shown in Figure 10 on the right, form when the meniscus has enough inertia to pinch off into a satellite drop. If the drop is small enough, the drop can return to the meniscus and backward-merge with it. Meniscus satellites are considered more severe than ligament satellites because of its difficulty to track the placement of such a drop that can be carried by air current. These satellite faults can lead to inaccurate placement on a substrate [15].

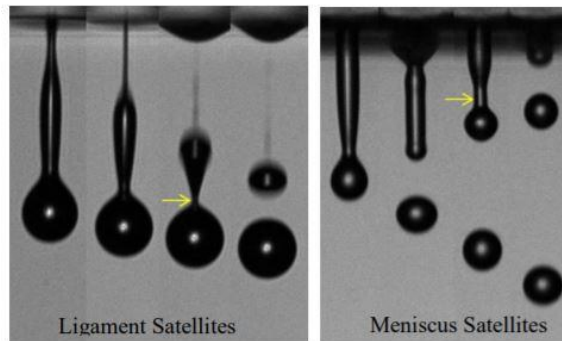


Figure 10. Ligament satellite (left) and meniscus satellite (right) accompanied with arrows to show pinch-off points [15]

Other faults that may occur during jetting are contamination and bubble formation. Contamination results from dust in the air that may form films at the surface of the orifice or inside the fluid channel. Bubble formation can occur when bubbles are aspirated in the fluid channel and require removal via flushing the system [15]. Working in a clean environment to avoid contamination and making sure that the hydraulic lines are not in need of maintenance are both critical to have consistent results for generating desired droplet characteristics.

2.5 Current Methods for Calibrating a PZT Single Nozzle Inkjet

Piezo inkjets are able to dispense different fluids with unique rheological properties. Inks can range from common solvents to experimental inks. But when a new ink type is introduced to the inkjet system, an optimal waveform must be generated to dispense a droplet. Waveform tuning is the calibration process used to narrow down the optimal waveform. The basic idea behind this calibration is to derive an actuation waveform that will generate the droplet characteristics that a user desires. Calibrating a piezo inkjet can be achieved using automated or manual waveform tuning.

Automatic piezo-jet waveform tuning is achieved using global optimization techniques, where various types of waveforms are used to generate several droplets until the best waveform is narrowed down to produce the desired characteristics. One of these optimization techniques is called a genetic algorithm. Genetic algorithms (GA) are random-based classical evolutionary algorithms, where random changes are applied to current solutions to generate a new one [18]. For inkjets, a solution is a parametric actuation waveform such as unipolar, bipolar, and tripolar waveforms. GAs integrated with computer-vision-based monitoring of in-flight drops creates a “machine-in-loop” structure that can evaluate a waveform solution based off key performance metrics estimated from the real-time images of jetted drops [15]. Figure 11 demonstrates the process used to find the optimal solution.

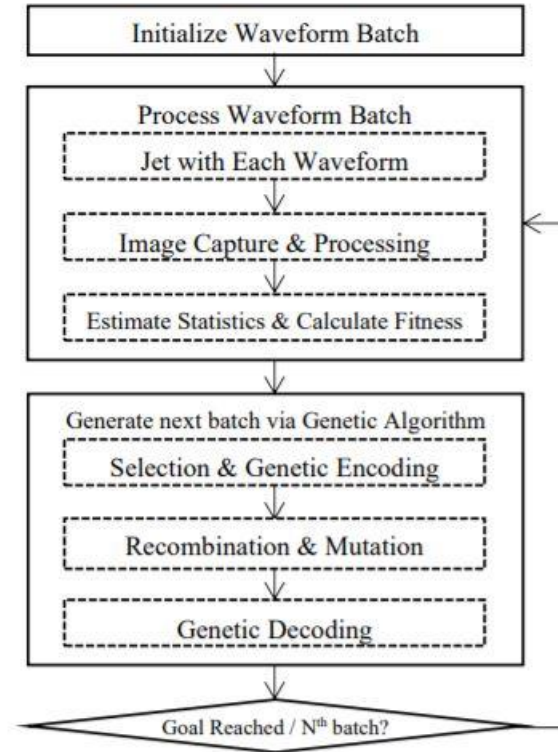


Figure 11. Block diagram describing genetic algorithm sequence for inkjet calibration [15]

The calibrating sequence in Figure 11 was created using a MicroFab piezo inkjet and associated hardware and software equipment from the experimental setup used in previous work. The waveform parameters, such as dwell time and peak voltage, are inputted to a JetDrive III controller to produce an actuation waveform that causes an ejection of drops that are captured using in-flight imaging and processed to determine their characteristics, such as drop volume and drop velocity. These droplet characteristics are used to derive a fitness metric to characterize the performance of each waveform. Once the waveform is tested, the GA identified waveforms based on their fitness to recombine and mutate into new waveforms [15]. Waveforms are tested until the optimal solution is produced to achieve the desired performance goal of the droplets.

Manual waveform tuning is similarly an iterative process but requires a user's constant input and observation. A user must first select the type of actuation waveform profile and decide its initial waveform parameters using the JetDrive controller. Then the generated droplets are

observed using an in-flight imaging system. If the droplet has any evident physical deformities, then the user must adjust key waveform parameters such as the dwell time, dwell voltage, and rise or fall times. Once the waveform tuning is completed, the droplets are observed again to determine if the adjustment helped get rid of any droplet deformities or made them worse. The cycle continues in tweaking the waveform parameters and observing the outputted drops until the user is satisfied with the results.

This research project focused on determining the jettability of nonpolar liquids using manual waveform tuning. The experimental ink may just pulsate at the orifice without ejecting any droplets or can inversely dispense a continuous stream of ink rather than a frequency of drops. Manually tuning individual parameters was conducted to gain insight into threshold values at which the two extreme jet types occur. The results can benefit projects moving forward that experiment with automatic waveform tuning since the algorithms must have preliminary data to effectively start the tuning process. In particular, manual tuning can be a good first approach to generate viable parents for a GA.

Chapter 3: Experimental Investigation

3.1 Experimental Setup

To understand how a given actuation waveform affects drop properties, a reliable imaging processing setup up is required to capture in-flight images of these ejected drops for analysis. In-flight images can help determine visually if droplets are being dispensed at all and if they have any physical deformities. Once proper spherical droplets are produced, these images can then be analyzed to calculate the sought-after drop volume. Figure 12 demonstrates the experimental setup used for the image acquisition of the jetted droplets.

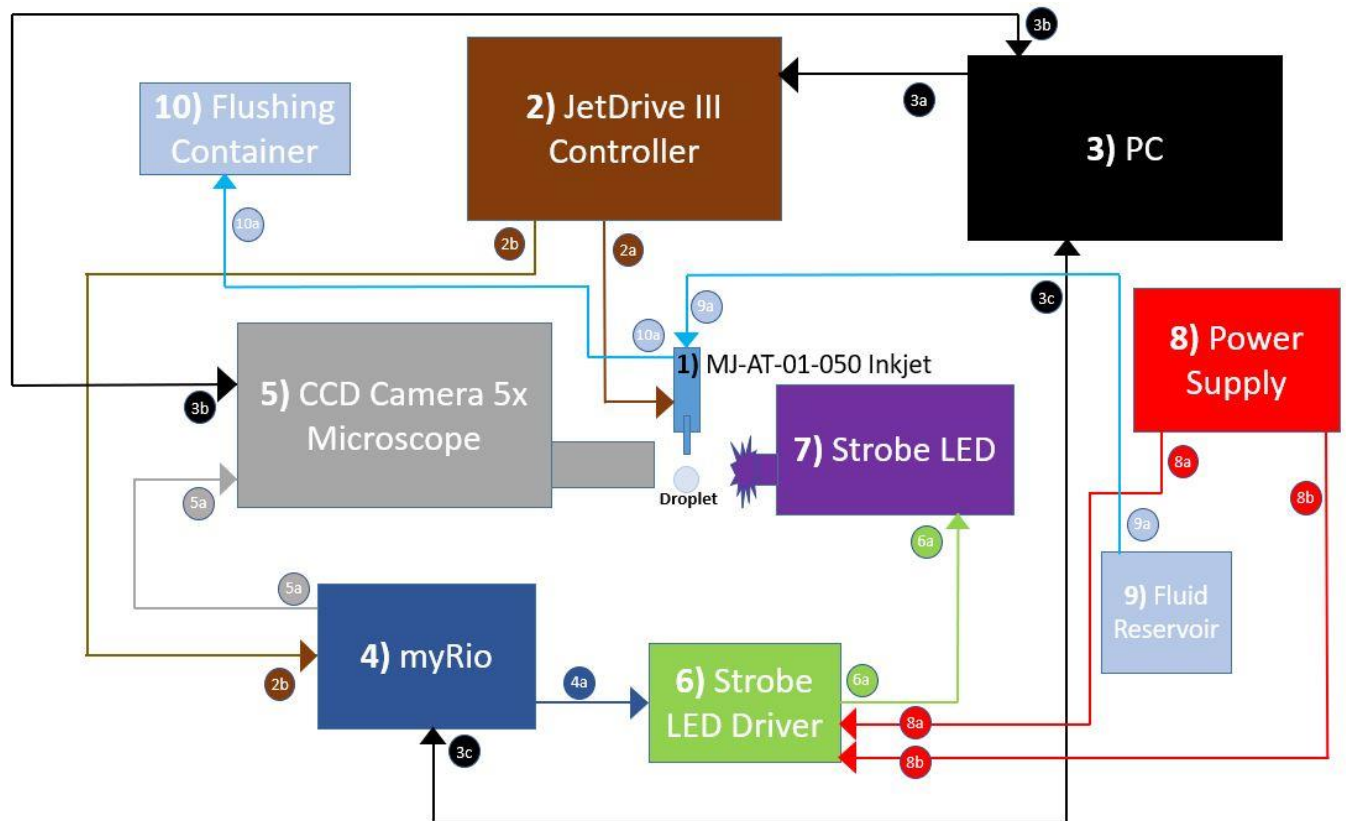
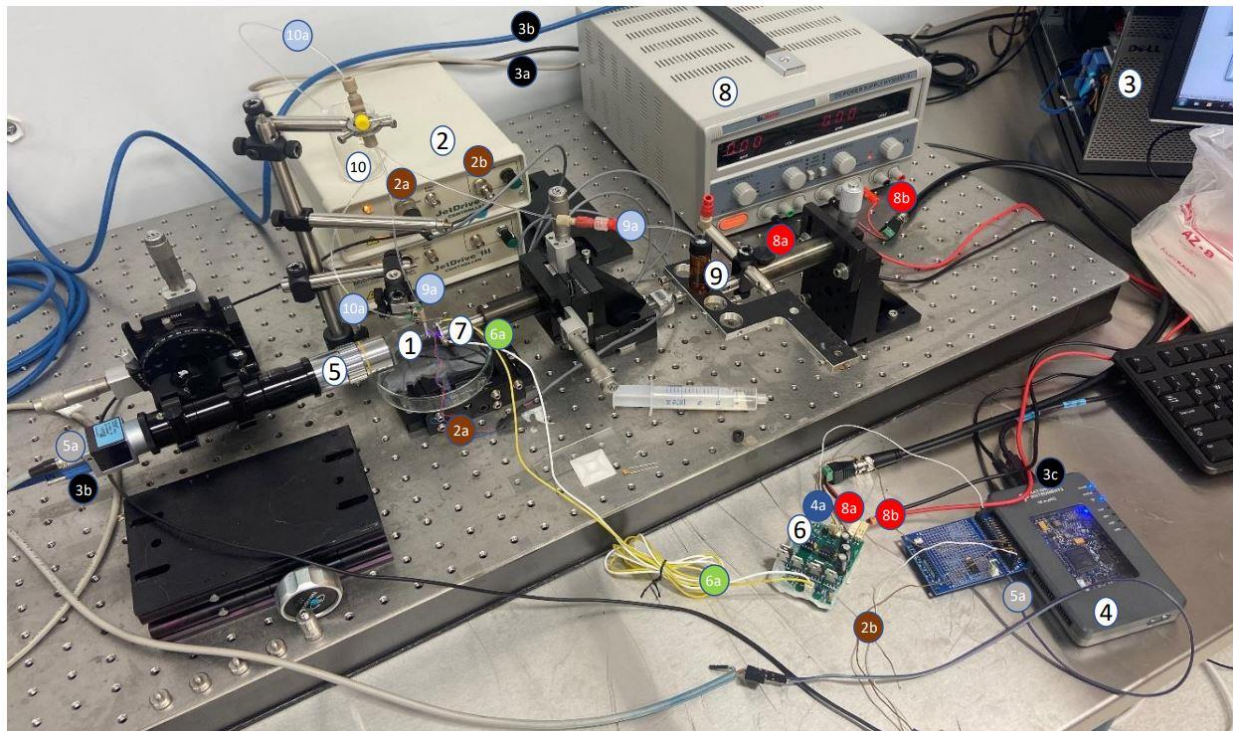


Figure 12. Single-nozzle piezo inkjet experimental setup diagram for testbed to acquire in-flight images

The boxes are labeled with the components that were used in the experimental setup. The alphanumeric labels on the arrowed-lines demonstrate the wires, cables, or hydraulic tubing that link these numbered components together. If the lines are one-directional then that is the path that information or ink flows, and if they are bidirectional then information is fed both ways. Figure 13 below shows a picture of the physical experimental setup to exhibit what the components and wiring looks like in the diagram above.



The single nozzle piezo inkjet used in this setup is a MicroFab MF-AT-01-050 dispenser (labeled as (1) in Figure 13). An annular PZT element, poled radially, is bonded to the exterior of a glass tube whose inner diameter is tapered at one end to create an orifice with a diameter of 50 μm . Whenever a voltage differential is applied to the annular PZT, the ring-shaped actuator will either expand or contract radially depending on the electrical field generated between the inner and outer electrodes. The actuator's expansion or contraction is transmitted to the glass tube and generates negative and positive pressures inside the chamber to create acoustic waves that cause fluid to emerge from the orifice. The dispensed ink forms into a spherical drop by surface tension forces. The piezo inkjet draws its jetting fluid from a fluid supply reservoir (labeled as (9) in Figure 13). The fluid reservoir is filled with a desired experimental ink, such as water and toluene for this project. The reservoir is pressurized and when the system is primed, the fluid runs through the hydraulic tubing from the reservoir to the jet and from the jet to a flushing container (labeled as (9a), (10a) and (10) in Figure 13). This tubing material was selected to withstand the corrosiveness of some nonpolar solvents, such as toluene, xylene and ethyl acetate. Whenever new inks are introduced, air bubbles or contaminants have a possibility of being present. As a syringe provides pressure from the reservoir's side of the system, the system can be purged, and the flushing container is used to capture the excess fluid. Once any air bubbles or contaminants are eliminated, the system is ready generate actuation waveforms to the PZT element to jet the fluid properly.

The piezo inkjet receives voltage actuation waveforms through small gauge wires soldered to the inner and outer electrodes.

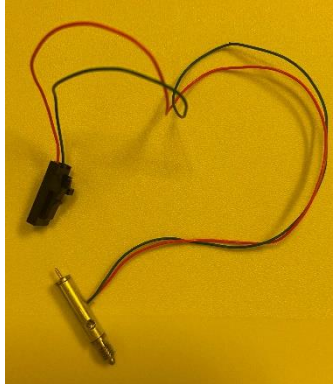


Figure 14. MJ-AT device with wires and connector

Figure 14 shows how these wires are connected to the piezo inkjet, the wires are twisted and placed together in a connector that matches to the output cable of a MicroFab JetDrive electronics box that generates custom waveforms.



Figure 15. The front pane of a MicroFab JetDrive III Controller

The MicroFab JetDrive III Controller is designed to provide complex actuation waveforms to MJ microdispensing devices (labeled as (2) in Figure 13). The cable that connects to the piezo inkjet's wire connector is used to provide it these waveforms (labeled as (2a) in Figure 13), while another cable is connected using an RS232 serial connector to the PC (labeled as (3a) connected to (3) in Figure 13). The JetDrive controller is computer-controlled by a JetServer4 software program that is run on the PC, this software allows the user to create custom complex actuation waveforms by tuning specific waveform parameters.

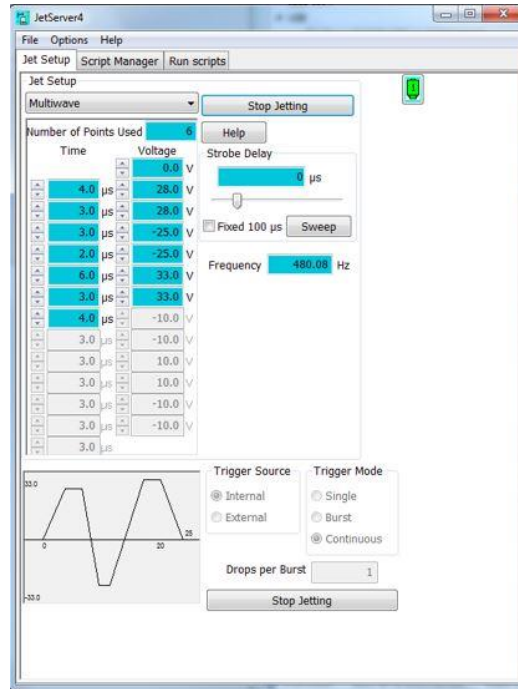


Figure 16. JetServer4 software program to create custom actuation waveforms generated by JetDrive III

A user must first select a waveform type, which is typically standard or multiwave. A standard waveform type has a bipolar profile while a multiwave type can generate unipolar or tripolar profiles by setting the number of points used to 2 or 6 respectively. A unipolar waveform is defined by four parameters, which are the rise time, dwell time, fall time and dwell voltage. A bipolar waveform is defined by seven parameters and a tripolar waveform is defined by ten parameters. A user can get creative by adjusting the waveform parameters. In Figure 16 for example, a tripolar waveform is selected and the parameters that are set manually are the rise times, dwell and echo times, fall times, and dwell and echo voltages. The JetDrive controller receives the custom waveform from the jetting software and the actuation waveform is directly applied to the PZT element to generate droplets.

To capture real-time images of in-flight droplets, a NI myRIO (NI, Inc., Austin, TX) device is the core unit to make this possible due to its data acquisition and controller capabilities (labeled as (4) in Figure 13). The JetDrive controller sends an external trigger to the myRIO for

real-time control of a camera and an LED strobe output/delay. This actuation trigger can be thought of as a master trigger, an internal clock with a certain number of ticks. The camera exposure can be regulated based on a specified number of timesteps, the camera used is a Basler acA1300-30gm through a Mitutoyo microscope with a 10x apochromatic lens (labeled as (5) in Figure 13). The myRIO also uses the actuation trigger to produce a programmable timed illumination trigger pulse for a high brightness strobe LED that is driven by an amplifier circuit (labeled as (7) and (6) respectively in Figure 13). The master trigger signals the start of the dispense of one drop, the camera enabled can capture images of the dispensed droplet by the contrast created from the illumination of the strobe of the LED. If the LED strobe is delayed incrementally during the capture sequence, the varying drop position in-flight can be captured depending on different strobe delay times and strobe pulse times [15]. These parameters, such as strobe delay, strobe pulse time, and camera exposure and reset time, are all configurable through the myRIO host LabVIEW virtual instrument (VI) program. The circuit diagram below is for the UCC27524 5A High Speed Low-Side Power MOSFET Driver.

Figure 17. Circuit diagram and PCB used for the amplification of a strobe LED [15]

This system creates clear and reliable image acquisition due to the synchronization of the camera and strobe LED with the jetted drops using the myRIO FPGA program. More detail about this experimental setup can be found in the work by Snyder [15].

3.2 Replicating Results Jetting Water

Before investigating the jettability of toluene with the single nozzle piezo inkjet, known results of jetted water droplets from previous work were attempted to be replicated. Although the results presented by Snyder were obtained using automated waveform tuning techniques [15], this work implemented the waveform parameters manually due to software constraints at the time of experimental development. This process involved manually adjusting the waveform parameters in the JetServer4 software and observing the dispensed droplet, and the continuous iteration between adjustment and observation continued until the generated droplet was a proper spherical shape without any physical deformities. Pre-experiment checkups and maintenance were performed to ensure proper jetting behavior, as explained in the following sections.

3.2.1 Troubleshooting Software and Hardware Issues

One of the issues encountered was strobe LED functionality and reliability. These issues, such as the LED not turning on were often related to faulty or loose wiring. Since the connector for the strobe controller is quite robust, unplugging and reconnecting the connector could potentially resolve the issue, but more often the issue lied on the LED-side wiring. The wires were manually soldered to the positive and negative connection pads of the LED board, making them susceptible to user error creating loose connections at the soldered ends. Thus, swapping out the LED and/or resoldering the wires properly usually took care of this issue. If the issue was with LED strobe event, for example the LED stuck in a high-brightness “always-on” mode, as

shown in Figure 18, it often meant a short in the electrical connection. Upon diagnosing this situation, the cause of this issue often came from improper electrical isolation.

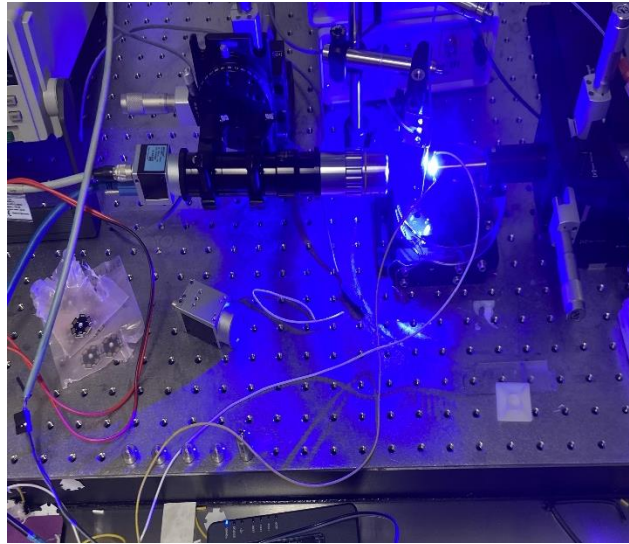


Figure 18. Example of LED shining at an extremely high brightness level

Whenever the imaging system was booted up, proper alignment between the camera, the strobe LED and the inkjet nozzle had to be attained. This was difficult due to high magnification of the camera objective. Alignment issues could be overcome by taking the LED off of the rod it is fixed to by an adhesive and moving it closer in front of the camera until the screen displaying the camera's image goes from pitch black to fully white. This allows the user to incrementally adjust the camera position by following the line of sight with the light of the LED until the nozzle orifice is found. The problem with this method is that as the LED is placed back on the rod fixture, the exposed wire or soldered end of the LED could make contact with the metal rod, shorting the system and causing an op amp in the LED driver circuit to explode.

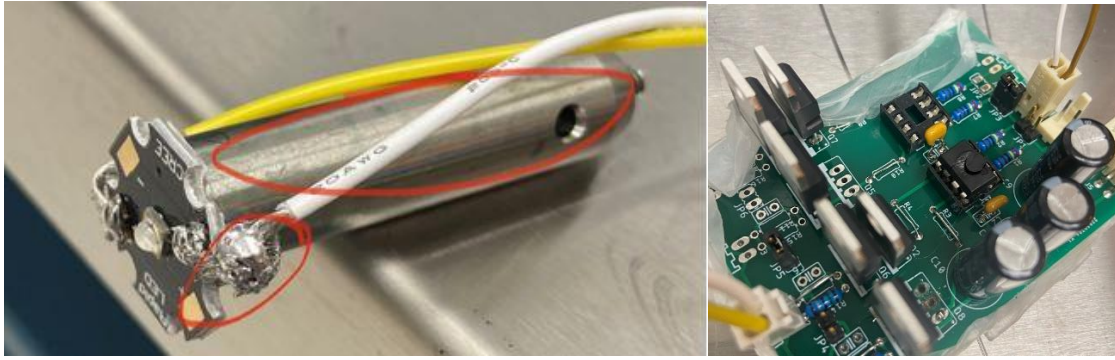


Figure 19. Two red circles show the exposed solder and wire of the LED along with the metal rod fixture the LED is attached to, along with the resulting blown op amp in the amplifier circuit

Figure 19 shows the potential contact points mentioned above and an example of an affected op amp. with the metal rod, shorting the system and causing an op amp in the LED driver circuit to explode. Replacing the op amp in the circuit would resolve this issue. To avoid this potential short, the power supply should remain off when adjusting the LED near the metal rod fixture. The tape shown in Figure 18 is used to separate the soldered points under the PCB from the metallic surface of the workbench to avoid a short as well.

Another setback that had to be overcome was clogging at the nozzle orifice of the piezo jet. Water droplets were unable to eject using any type of actuation waveform, even when waveform parameters were adjusted to settings successfully applied in previous work. The camera was used to identify this issue, since the image acquisition showed the nozzle clearly but not any jetted drops due to debris blockage at the orifice. An oscilloscope was used to verify signals were properly sent to the myRIO, to the LED driver, and to the piezo jet. Then to confirm whether the inkjet was jetting any fluid, a unipolar actuation waveform was applied and the jet was left running for an hour, but no water buildup was found. If the jet was working properly, a small puddle of water would have formed from the dispensed droplets. A silicon wafer was placed under the jet for this hour to make it easier on the user to determine if any residual water was dispensed or not.

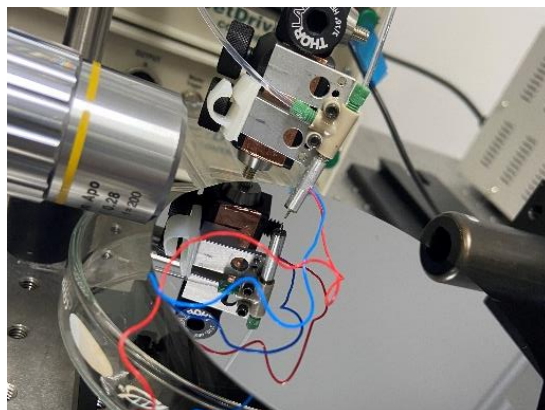


Figure 20. A silicon wafer placed less than 1 mm away from the nozzle head to observe any puddle formation from dispensed water droplets

Figure 20 shows how clear and reflective the surface of a silicon appears. If any grouping of water droplets formed on this surface it would be simple to distinguish. After confirming the inkjet did not dispense any fluid, the next step was to purge the inkjet. Purging the inkjet system requires the user to connect a syringe to the hydraulic tubing and force the fluid out of the jet nozzle to get rid of any potential contaminants.

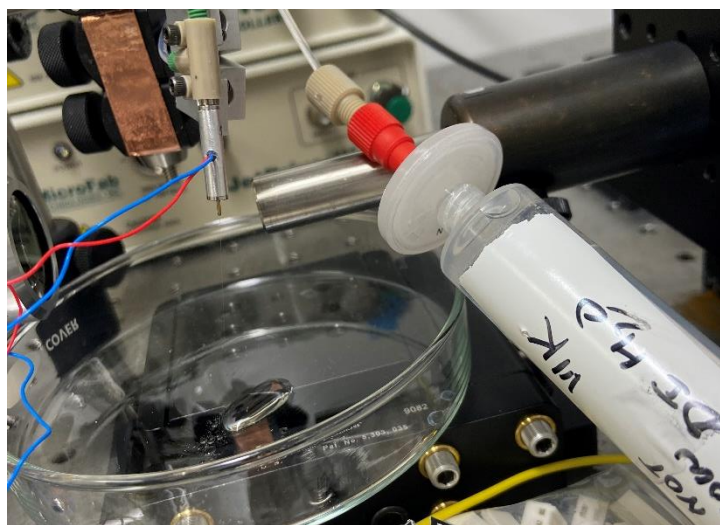


Figure 21. Syringe placement with a nano-filtration system to purge water through the system and force water out of the inkjet orifice

Figure 21 shows the result of purging the system, a continuous stream of water poured out of the nozzle orifice and the puddle of water formed in the petri dish. After purging the

inkjet, air bubbles appeared to show up in the hydraulic tubing of the system. To eliminate these air bubbles, a syringe was used to pressurize the fluid reservoir and a yellow lever served as a valve that was turned to redirect fluid flow and release any remaining air or contaminated fluid into the flushing container. This process is shown in the figure below.

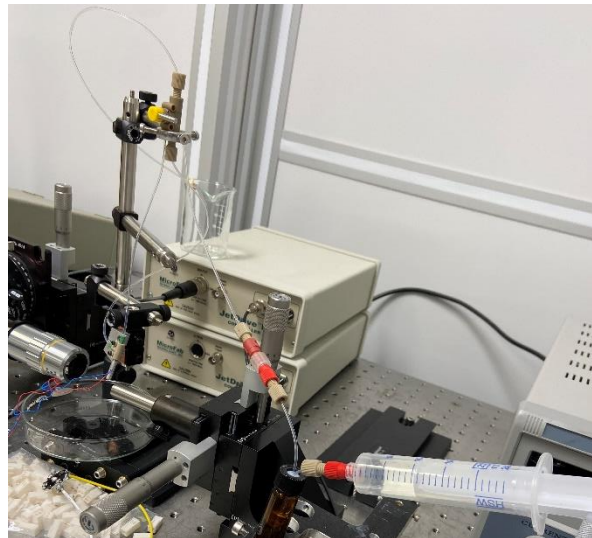


Figure 22. Syringe pushing air into the fluid reservoir to pressurize the system and eliminate air bubbles by turning the yellow lever to open a valve and release excess into the beaker

If after these steps were taken, the inkjet was still unable to jet any fluid, the last resort was to assume the piezo element was no longer working and a replacement jet was ordered from MicroFab Inc. A brand-new piezo inkjet was attached and precautions were taken by making sure the threaded fitting was tightened properly because if any loose contact was present the jetted fluid would be trapped in the connector. Even with the new device implemented into the system, the jet was purged once more to ensure the nozzle had a clear pathway. Then the JetDrive controller provided a unipolar actuation waveform, and in less than 5 minutes of tuning the drops started to dispense appropriately. This was verified by puddle formation on the silicon wafer placed below the inkjet nozzle and by using the image capture system to observe the generated droplets.

Once the necessary troubleshooting was completed and the entire inkjet system was working correctly, the focus was able to be shifted towards experimenting with the waveform parameter settings. Through manual waveform tuning, a user can work towards generating droplets with a specific drop volume in mind. Previous work has provided known drop volumes of droplets generated by unipolar, bipolar and tripolar actuation waveforms [15]. Those results are attempted to be replicated using manual tuning techniques.

3.2.2 Replicating Known Drop Volumes Using Manual Waveform Tuning

The manual waveform tuning process was performed with a basic unipolar actuation waveform arbitrarily to understand how changing the rise time, dwell time, fall time, and dwell voltage affected the generated drop formation. Insight was gained by: a) taking note of how the jetting fluid would dispense with drop deformations, b) how the fluid dispensed the optimized spherical drops, and c) whether the jetted fluid either pulsated at the nozzle orifice or ejected in a continuous stream. But to replicate the known drop volumes, previous work provided the settings for the waveform parameters used for unipolar, bipolar, and tripolar waveforms [15]. Those known waveform parameters were applied as a baseline, and the response was observed. The tripolar waveform was the only profile that generated optimal drops based off the known waveform parameter settings. The known unipolar and bipolar shapes resulted to be non-jetting waveforms. Non-jetting waveforms are those that do not generate acoustic pressures sufficient to drive drops out of the nozzle. A negative acoustic pressure generated during the rise time of the waveform retracts the meniscus, and as the negative pressure wave reflects at the reservoir it becomes positive and positive pressure is amplified by the de-charging of the PZT element. This situation is known as pull-push action, and if the resulting positive pressure cannot drive the ink to overcome inertia, viscosity, and surface tension forces, drop formation does not occur [8].

Parameters needed to be adjusted further for the non-jetting waveforms in order to properly dispense drops. The real-time imaging system was used to inspect the generated drops and parameters were adjusted until the droplet was as close to a spherical shape as possible. Once the inkjet was consistently jetting acceptable shaped droplets, a sample size of 10 drop images were captured for each actuation waveform profile. Those 10 images were saved and later analyzed using the ImageJ image processing and analysis tool. To calculate the desired drop volume characteristic, the main assumption made from these images were that the droplets adhered to a perfect spherical shape due to surface tensions in-flight. The basic formula of a sphere was used to calculate the drop volumes.

$$V = \frac{4}{3} \pi r^3 \quad (1)$$

This equation was used due to image analysis being done in ImageJ, the user can easily and consistently obtain the drop diameter, but inconsistencies were met when calculating the surface area of the droplet as the shading was not uniform across the entire drop. Figure 23 below shows how a dispensed droplet may appear, and the surface area calculation may have omitted the white center of the drop with the shaded outline.

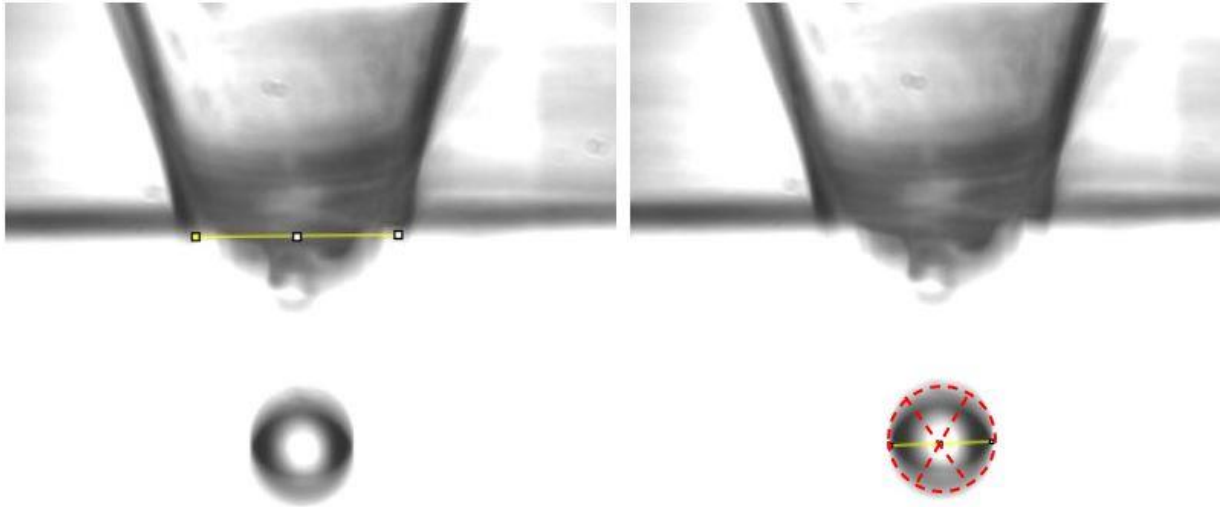


Figure 23. ImageJ drop analysis; the left panel shows how to set the scale of the image with the orifice distance and the right panel shows the line feature measuring the diameter of the drop along with a red outline of the next lines a user would measure

In Figure 23, the first step of this analysis was setting the appropriate scale for the entire image with the known distance of the nozzle orifice. The distance of pixels for that line was set to the known distance of 50 μm . The line feature was then used to evaluate the diameter of the drop based off the set scale. To account for potential user error, the line feature was used 10 times for each drop image in case the user may begin or end the line feature a few pixels too short or too long. The red dashed circle outline in Figure 23 shows how the user should imagine the droplet was conformed to, and 10 drop diameters should be determined at different angles to calculate an average diameter. Once the data was collected, the drop volumes were calculated for the jetted water droplets for each actuation waveform type.

Table 1 below shows the data collected for the jetted water drops resulting from use of a unipolar actuation waveform.

Water: Unipolar Waveform													
Droplets	Drop Diameter Measurement (um)										Calculations		
Drop #	1	2	3	4	5	6	7	8	9	10	Avg. Drop Diameter (um)	Drop Volume (um^3)	Drop Volume (pL)
1	27.6010	27.0790	27.2880	26.5940	27.3240	27.3880	26.7990	27.8660	27.7240	28.5110	27.4174	10791.3910	10.7914
2	28.3170	28.0960	28.7620	27.4510	27.9020	29.1510	27.9720	28.6570	28.4440	28.9510	28.3703	11956.1239	11.9561
3	29.4380	28.7930	28.7790	29.3270	28.6460	28.4800	27.9870	29.6500	28.7660	28.7810	28.8647	12592.1465	12.5921
4	28.2000	27.6200	27.8340	27.6210	27.5510	28.6930	28.7580	28.4940	27.5040	27.7190	27.9994	11493.3014	11.4933
5	28.4060	28.7760	28.4380	28.6200	29.1810	28.9600	27.9980	27.7370	27.7400	28.4170	28.4273	12028.3335	12.0283
6	29.2290	28.8840	28.4700	29.1240	28.0040	27.9630	28.5680	28.8960	28.6330	28.8500	28.6621	12328.8521	12.3289
7	28.5360	27.0930	28.5280	27.3620	27.9630	27.2660	27.5160	27.9670	28.5690	27.7420	27.8542	11315.4203	11.3154
8	29.9150	28.7710	27.9120	29.0560	28.5530	27.8530	28.2950	27.4140	28.8850	27.9160	28.4570	12066.0734	12.0661
9	29.6800	29.6960	28.2660	28.6810	30.0160	28.5770	27.9870	29.8980	28.8280	28.8860	29.0515	12838.2050	12.8382
10	28.5720	29.1450	28.9930	28.8740	28.8420	29.2970	28.3390	29.1770	28.1100	28.3380	28.7687	12466.9247	12.4669
												Average Drop Volume (pL)	11.9877
												Drop Volume St. Dev.	0.6299

Table 1. Drop diameter data collected and drop volume calculated for a unipolar actuation waveform

For each of the 10 drop images, 10 measurements of the drop diameter were obtained in the ImageJ analysis. The average drop diameter was used to calculate the drop volume and the volume unit was converted to the standard picolitre. The drop volumes from all 10 images were then averaged as well, and the average drop volume was found for this set of data.

Table 2 below shows the waveform parameters and the VI settings for the camera and strobe LED, these settings produced the generated droplets with the image clarity provided from the display.

Unipolar Actuation Waveform Settings				
JetServer4 Parameters		SAMBA RT Main.vi Settings		Picture of Generated Drop
# of Points Used	2	Strobe Up Delay Tics	200	
Rise Time (us)	2	Strobe Pulse Width Tics	6	
Dwell Time (us)	18	Camera Reset Time (tics)	620000	
Fall Time (us)	6	Walk Step (tics)	30	
Dwell Voltage (V)	31	N Steps	300	

Table 2. Unipolar actuation waveform parameters and the LabVIEW VI settings used to generate the water droplet images

The JetServer4 software also provided an image of what the unipolar profile looked like as the corresponding parameters were changed, and once the parameters were optimized the

generated droplet was captured clearly by the camera. In the table below, these waveform parameter settings and the evaluated drop volume are compared to the known results from previous work [15].

Unipolar					
	Rise Time (us)	Dwell Time (us)	Fall Time (us)	Dwell Voltage (V)	Drop Volume (pL)
Previous Work	6	9	14	30	29.74
Tuned Results	2	18	6	31	11.99

Table 3. Comparison of previous work's unipolar waveform parameters and drop volumes to this project's results

The dwell voltage was similar, but the rest of the parameters and resulting drop volumes were quite different. An interesting takeaway after performing the manual waveform tuning is how the dwell time was doubled, but the drop volume was less than half of the previous finding.

Table 4 below shows the data collected for the jetted water drops experiencing a bipolar actuation waveform.

Water: Bipolar Waveform													
Droplets	Drop Diameter Measurement (um)										Calculations		
Drop #	1	2	3	4	5	6	7	8	9	10	Avg. Drop Diameter (um)	Drop Volume (um^3)	Drop Volume (pL)
1	27.311	27.821	27.109	27.569	26.311	27.517	28.363	27.224	27.512	26.699	27.3436	10704.48306	10.70448306
2	27.857	27.518	26.927	26.947	26.575	26.128	26.075	26.447	27.292	26.762	26.8528	10138.35173	10.13835173
3	29.175	29.528	28.178	28.733	29.324	28.255	28.193	28.457	29.409	28.398	28.765	12462.11508	12.46211508
4	25.16	25.492	26.173	25.83	27.285	25.767	26.179	26.165	26.205	26.536	26.0792	9287.127698	9.287127698
5	28.285	27.408	26.534	26.759	27.98	28.583	26.127	28.255	27.408	27.505	27.4844	10870.69733	10.87069733
6	25.941	26.109	26.436	27.633	26.806	27.084	27.203	27.071	27.302	26.088	26.7673	10041.81744	10.04181744
7	26.642	25.905	26.071	25.248	26.436	25.137	26.322	26.058	26.021	25.724	25.9564	9156.55265	9.15655265
8	24.089	24.805	26.378	25.158	26.104	25.139	25.109	26.282	26.177	26.92	25.6161	8801.114118	8.801114118
9	23.994	25.787	26.895	25.516	25.158	25.259	26.657	25.096	26.42	24.293	25.5075	8689.650478	8.689650478
10	27.441	27.183	26.431	26.696	25.649	27.116	25.209	27.736	26.86	26.891	26.7212	9990.023162	9.990023162
												Average Drop Volume (pL)	10.0141933
												Drop Volume St. Dev.	1.14062991

Table 4. Drop diameter data collected and drop volume calculated for a bipolar actuation waveform

The JetDrive and VI settings are provided below along with the table comparing the findings from this work against published results [15].

Bipolar Actuation Waveform Settings				
JetServer4 Parameters (Standard Wave)		SAMBA RT Main.vi Settings		Picture of Generated Drop
Rise Time 1 (us)	2	Strobe Up Delay Tics	200	
Dwell Time (us)	11	Strobe Pulse Width Tics	7	
Fall Time (us)	11	Camera Reset Time (tics)	600000	
Echo Time (us)	7	Walk Step (tics)	20	
Rise Time 2 (us)	3	N Steps	200	
Dwell Voltage (V)	22			
Echo Voltage (V)	-18			

Table 5. Bipolar actuation waveform parameters and the LabVIEW VI settings used to generate the water droplet images

Bipolar								
	Rise Time 1 (us)	Dwell Time (us)	Fall Time (us)	Echo Time (us)	Rise Time 2 (us)	Dwell Voltage (V)	Echo Voltage (V)	Drop Volume (pL)
Previous Work	2	11	11	3	3	14	-12	18.3
Tuned Results	2	11	11	7	3	22	-18	10.01

Table 6. Comparison of previous work's unipolar waveform parameters and drop volumes to this project's results

The bipolar waveform settings used as the initial test waveform was derived from parameters based off known results [15], but droplets were not dispensed right away. Some parameters required a bit of tweaking to generate spherical droplets, a longer duration for the echo time was accounted for, and the peak voltages for both pulses were altered. The resulting drop volume was .55x the volume of the previous finding [15].

Table 7 below shows the data collected for the jetted water drops experiencing a tripolar actuation waveform.

Water: Tripolar Waveform													
Droplets	Drop Diameter Measurement (um)										Calculations		
Drop #	1	2	3	4	5	6	7	8	9	10	Avg. Drop Diameter (um)	Drop Volume (um^3)	Drop Volume (pL)
1	23.639	24.135	23.766	23.794	23.73	23.481	23.275	23.509	23.308	23.641	23.6278	6906.666415	6.906666415
2	24.013	23.618	23.685	23.154	23.329	23.475	23.082	23.566	23.548	23.384	23.4854	6782.542232	6.782542232
3	24.034	23.247	23.124	23.378	23.071	22.453	23.231	22.995	23.816	23.144	23.2493	6580.035956	6.580035956
4	24.534	23.08	23.636	24.187	24.188	23.573	23.887	24.311	24.07	23.057	23.8523	7105.414392	7.105414392
5	24.532	23.694	24.086	23.99	23.421	24.37	23.618	23.835	23.284	24.027	23.8857	7135.305008	7.135305008
6	23.635	23.516	23.031	22.219	22.786	23.522	23.703	23.511	23.756	23.869	23.3548	6670.019136	6.670019136
7	23.298	23.175	22.457	23.065	23.525	23.271	23.253	23.083	23.772	23.675	23.2574	6586.915759	6.586915759
8	23.53	24.717	24.211	23.855	24.284	23.535	23.072	23.443	24.336	24.05	23.9033	7151.089422	7.151089422
9	22.991	23.966	22.626	23.379	23.662	23.101	23.095	23.362	23.114	23.453	23.2749	6601.795898	6.601795898
10	23.636	23.671	23.36	24.416	24.397	24.389	24.059	23.449	23.789	24.019	23.9185	7164.740135	7.164740135
												Average Drop Volume (pL)	6.86845244
												Drop Volume St. Dev.	0.25315798

Table 7. Drop diameter data collected and drop volume calculated for a tripolar actuation waveform

The JetDrive and VI settings are provided below along with the table comparing this project's findings to the known results.

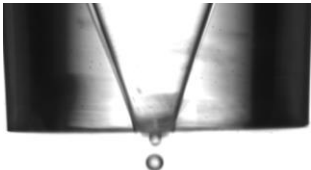
Tripolar Actuation Waveform Settings				
JetServer4 Parameters (Multiwave)		SAMBA RT Main.vi Settings		Picture of Generated Drop 
# of Points Used	6	Strobe Up Delay Tics	300	
Rise Time 1 (us)	4	Strobe Pulse Width Tics	9	
Dwell Time 1 (us)	3	Camera Reset Time (tics)	600000	
Fall Time 1 (us)	3	Walk Step (tics)	20	
Echo Time (us)	2	N Steps	200	
Rise Time 2 (us)	6			
Dwell Time 2 (us)	3			
Fall Time 2 (us)	4			
Dwell Voltage 1 (V)	28			
Echo Voltage (V)	-25			
Dwell Voltage 2 (V)	33			

Table 8. Tripolar actuation waveform parameters and the LabVIEW VI settings used to generate the water droplet images

Tripolar											
	Rise Time 1 (us)	Dwell Time (us)	Fall Time 1 (us)	Echo Time (us)	Rise Time 2 (us)	Dwell Time 2 (us)	Fall Time 2 (us)	Dwell Voltage 1 (V)	Echo Voltage (V)	Dwell Voltage 2 (V)	Drop Volume (pL)
Previous Work	4	3	3	2	6	3	4	28	-25	33	6.3
Tuned Results	4	3	3	2	6	3	4	28	-25	33	6.87

Table 9. Comparison of previous work's tripolar waveform parameters and drop volumes to this project's results

In this case, the known settings for the tripolar actuation waveform generated droplets that showed up very clearly on the image capture system with the expected spherical drop shape. No manual tuning was needed, and the percent difference between the expected and calculated drop volumes was only 9%. This low percent difference helps show how robust this higher order waveform is with regards to its consistent output across different inkjet devices.

It was reassuring finding that the tripolar actuation waveform results were similar, but the potential causes for the significant difference in drop volumes for unipolar and bipolar waveforms could be due to hardware variability or due to the tuning process. Previous work obtained the results using the automated tuning method by configuring the optimal waveform to minimize the drop volume while maximizing the drop velocity [15]. The drop velocity calculation that was tied with the automated tuning process was unavailable for this project. Although the unipolar and bipolar drop volumes were achieved to be smaller, the lack of consideration for a drop velocity could be a factor for the disparity. For the purposes of this project, investigating the jettability of nonpolar liquids, the studies only considered whether the inkjet could jet toluene or not. The image-capture system would confirm this through a user's observation and provide images to analyze as well.

Chapter 4: Results for Jetting a Nonpolar Liquid in a Piezo Inkjet

4.1 Toluene Background Information

The objective of this project was to determine the jettability of nonpolar liquids using piezo inkjets. There is an interest in jetting nonpolar solvents because of their similar properties to potential organic inks [4]. The droplets produced from piezo jets have excellent size and placement accuracy, and the behavior nonpolar solvents have with the substrates is sought after since they have desirable color reproducibility and adhesiveness to the substrate [3]. This project focused on toluene, an example of a nonpolar liquid.

4.2 Experimental Results for Toluene Jettability

The manual waveform tuning process was much more intricate when introducing toluene as the experimental ink. The safety and handling of toluene was of utmost importance, and before toluene was introduced to the hydraulic system, the system needed to be purged with isopropyl alcohol (IPA). Since the inkjet setup used water beforehand as a baseline, any residual water was swept away with the IPA purge. After purging the system with IPA, a syringe pushed air throughout the whole system to make sure the lines were dry. With the system cleaned, toluene was poured into the fluid reservoir, and the system was pressurized running toluene throughout the system. But this introduced the first setback shown in the figure below.

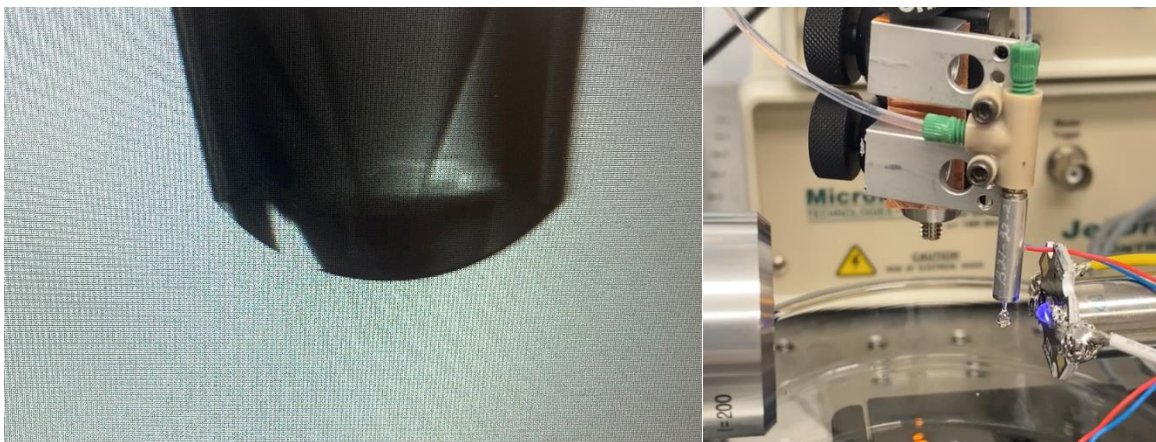


Figure 24. Toluene first introduced to the inkjet system and dripping out of the nozzle, the left shows the view through the camera and the right shows the drop visible to the naked eye

Even though the JetDrive controller was off, and no actuation waveform was applied to the PZT element, the toluene would leak out of the orifice and surface tension forces would lead to the fluid creeping up the outside of the nozzle. Due to toluene's unique rheology, if too much pressure was provided by the syringe the toluene would continue to leak. Waiting until the system stopped dripping on its own was an option but providing an actuation waveform to the jet would also help after swabbing the orifice with a cleaning wipe.

Once the system was in a stable state, the waveform parameter settings from the manual tuning process with water were applied to toluene as well. The unipolar waveform seemed to provide the closest results to eject continuous droplets, while the bipolar and tripolar waveforms only generated irregular streams of droplets with drop deformities and would spray toluene in random directions. Therefore, the manual waveform tuning focused on adjusting unipolar waveform parameters to generate spherical drops. With only a few parameters to configure, the attention converged on manipulating the dwell time and dwell voltage to different extremes and the resulting behavior of toluene at the nozzle orifice was consistent across these extremes. If the dwell parameters were too low the toluene would pulsate at the orifice without ejecting any fluid, but if the parameters were too high the toluene would flow out of the nozzle in a continuous

stream rather than a frequency of drops. Once the unipolar waveform parameters were tuned to the appropriate values, the applied actuation waveform started to generate consistent spherical droplets.

Unipolar Actuation Waveform Settings				
JetServer4 Parameters (Multiwave)		SAMBA RT Main.vi Settings		Picture of Generated Drop
# of Points Used	2	Strobe Up Delay Tics	200	
Rise Time (us)	11	Strobe Pulse Width Tics	10	
Dwell Time (us)	36	Camera Reset Time (tics)	600000	
Fall Time (us)	9	Walk Step (tics)	18	
Dwell Voltage (V)	60	N Steps	600	

Table 10. Unipolar actuation waveform parameters and the LabVIEW VI settings used to generate the toluene droplet images

Table 10 shows the settings used to achieve a consistent jetting of toluene drops. The rise and fall times were considerably high compared to results using water, and the dwell time and dwell voltage values were almost double the values used for the water baseline. Extra images of the toluene drops are included in the figure below to demonstrate that the images acquired were very clear.



Figure 25. A few of the toluene droplet images collected after the manual waveform tuning optimized a unipolar waveform that generated these nearly perfect spherical toluene drops

Ten images were collected again for the analysis in ImageJ to determine the volume of toluene drops. The table below shows the data collected for the jetted toluene drops experiencing the unipolar waveform that was manually optimized.

Toluene: Unipolar Waveform													
Droplets		Drop Diameter Measurement (um)									Calculations		
Drop #	1	2	3	4	5	6	7	8	9	10	Avg. Drop Diameter (um)	Drop Volume (um^3)	Drop Volume (pL)
1	37.492	38.598	38.511	37.854	38.265	37.048	38.983	38.537	38.295	38.732	38.2315	29259.21267	29.25921267
2	38.209	38.399	39.229	38.375	37.226	38.544	37.955	37.392	39.249	38.707	38.3285	29482.48541	29.48248541
3	38.537	37.182	37.629	37.888	38.171	37.895	39.532	38.106	38.545	39.631	38.3116	29443.50389	29.44350389
4	37.959	37.801	37.31	38.766	38.064	38.689	38.031	37.98	38.267	38.516	38.1383	29045.75128	29.04575128
5	37.909	38.955	38.121	38.788	38.283	39.471	39.5	38.67	37.909	39.321	38.6927	30330.93027	30.33093027
6	38.439	38.655	39.464	39.056	39.545	38.07	38.487	39.049	38.569	39.194	38.8528	30708.99402	30.70899402
7	38.662	38.947	38.606	38.639	39.253	39.258	38.311	39.341	38.893	38.21	38.812	30612.35143	30.61235143
8	38.525	38.239	38.831	38.283	39.127	39.155	39.19	38.519	38.626	37.873	38.6368	30199.66127	30.19966127
9	39.157	38.888	38.783	38.77	39.83	39.936	39.018	38.128	38.183	38.43	38.9123	30850.2954	30.8502954
10	40.006	39.344	39.283	39.698	39.26	40.64	39.37	39.436	38.239	38.695	39.3971	32017.79257	32.01779257
												Average Drop Volume (pL)	30.19509782
												Drop Volume St. Dev.	0.911515805

Table 11. Toluene drops diameter data collected and drop volume calculated for a unipolar actuation waveform

The average drop volume for toluene was 30.2 pL, a much greater volume than those found with the results from water. Overall, the optimized unipolar waveform produced from the manual tuning successfully generated consistent drops of toluene, which proved that nonpolar liquids can be jetted with piezo inkjets.

4.3 Additional Insights for Toluene Jettability

Attempts were made to repeat results from jetting of toluene but another obstacle was encountered. A contaminant appeared in the system that was not present in previous experiments. A possible explanation for this is that when the clean wipe was used to dry off the accumulation of toluene on the outer surface of the nozzle, a microfiber piece of the wipe could have been sucked into the inside of the inkjet orifice. This contaminant was potentially present because of the malpractice of trying to clean the nozzle while the JetDrive controller was actively sending actuation waveforms to the PZT element. The forces at the pulsating orifice pulled the contaminant inwards.

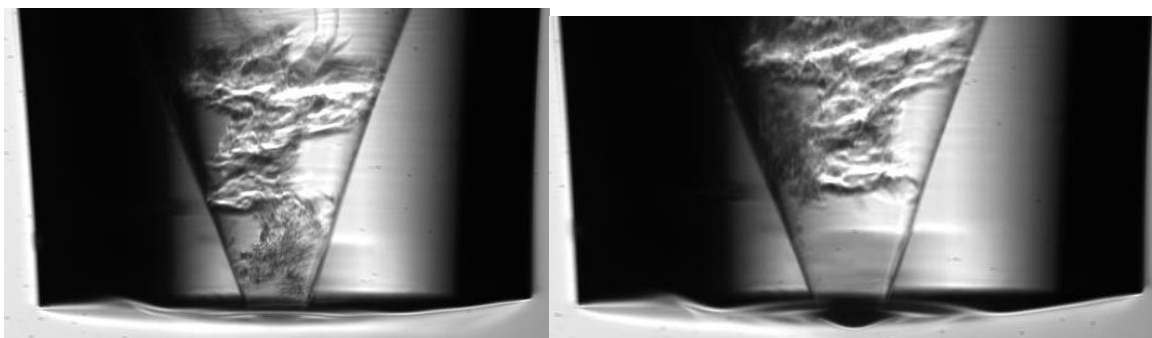


Figure 26. Images exemplifying a compromised inkjet system with a contaminant trapped in the nozzle

Figure 26 shows the stated of the contaminated system. The entire inkjet setup was purged with IPA three times, but the contaminant was persistent and could not be expelled from the system. Since the contaminant stayed within the system, another round of manual waveform tuning was necessary. Surprisingly, the inkjet was still able to jet drops of toluene as it did in its pre-contaminant state. The figure below shows the optimal unipolar waveform parameters that were produced for the inkjet to dispense toluene drops in its post-contaminant state.

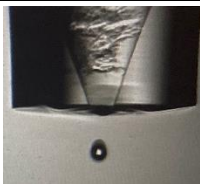
Unipolar Actuation Waveform Settings				
JetServer4 Parameters (Multiwave)		SAMBA RT Main.vi Settings		Picture of Generated Drop
# of Points Used	2	Strobe Up Delay Tics	300	
Rise Time (us)	3	Strobe Pulse Width Tics	18	
Dwell Time (us)	9	Camera Reset Time (tics)	550000	
Fall Time (us)	2	Walk Step (tics)	22	
Dwell Voltage (V)	54	N Steps	400	

Table 12. Unipolar actuation waveform parameters and the LabVIEW VI settings used to generate the toluene droplet images in the post-contaminant state

Another 10 images were captured, and the ImageJ analysis provided data for the table below.

Toluene: Unipolar Waveform														
Droplets		Drop Diameter Measurement (um)										Calculations		
Drop #	1	2	3	4	5	6	7	8	9	10	Avg. Drop Diameter (um)	Drop Volume (um^3)	Drop Volume (pL)	
1	35.135	35.703	35.48	35.81	35.204	35.915	35.819	36.567	35.54	35.742	35.6915	23806.361	23.806	
2	35.917	36.193	36.857	35.342	35.502	35.24	35.009	35.659	36.357	35.724	35.78	23983.890	23.984	
3	36.007	35.16	35.191	35.988	35.568	35.331	35.983	36.244	35.35	35.805	35.6627	23748.779	23.749	
4	34.702	34.311	35.81	35.461	34.511	34.696	35.707	35.064	36.009	35.461	35.1732	22784.225	22.784	
5	34.582	34.881	35.33	36.547	36.107	35.409	34.615	35.079	35.766	36.03	35.4346	23295.993	23.296	
6	34.972	35.549	35.53	36.463	36.658	34.66	36.612	34.781	35.283	34.674	35.5182	23461.267	23.461	
7	34.972	34.555	35.83	35.939	36.615	35.817	35.498	36.209	35.179	35.984	35.6598	23742.986	23.743	
8	34.855	35.719	34.255	35.304	35.879	35.797	34.892	35.5	35.797	34.707	35.2705	22973.834	22.974	
9	34.876	36.796	35.28	35.3	35.084	35.854	35.454	35.875	35.183	35.278	35.498	23421.261	23.421	
10	35.129	36.144	36.144	35.521	35.736	34.196	36.221	36.063	36.091	35.026	35.6271	23677.729	23.678	
												Average Drop Volume	23.490	
												Drop Volume St. Dev.	0.383	

Table 13. Toluene drops diameter data collected and drop volume calculated for a unipolar actuation waveform in the post-contaminant state

The average drop volume was 23.49 pL for the toluene drops in the post-contaminant state was less than the drop volume found in the pre-contaminant state. This insight warrants that even when a contaminant is present for this experiment, toluene droplets are still jettable with a piezo inkjet setup. If the PZT actuator is working properly and a nonpolar solvent is jettable, then the need for an automated waveform tuning process is much more beneficial as the user can still set the algorithm to generate desired droplet characteristics.

The effects of dwell time and voltage were also studied using the waveform parameter settings from Table 12 as a baseline, the dwell time or dwell voltage were each decreased and increased drastically while keeping the other constant to determine how the nonpolar solvent reacted.



Figure 27. This resulting image shows a drop pulsating back into the orifice when the dwell time was reduced from 9 μ s to 5 μ s while keeping the dwell voltage at 54 V

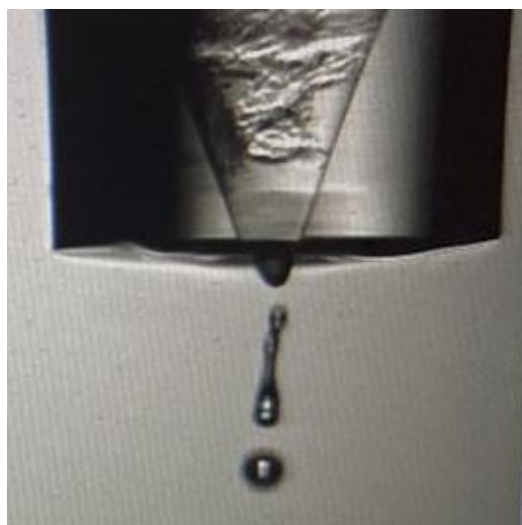


Figure 28. This resulting image shows a dispense drop followed by a ligament satellite in quick succession when the dwell time was increased from 9 μ s to 15 μ s while keeping the dwell voltage at 54 V

If the dwell time duration is below the optimal dwell time value, the toluene pulsates at the orifice without ejecting any fluid. But if the dwell time is greater than the optimal dwell time, then the toluene dispenses at irregular intervals and is experiencing satellite deformities.

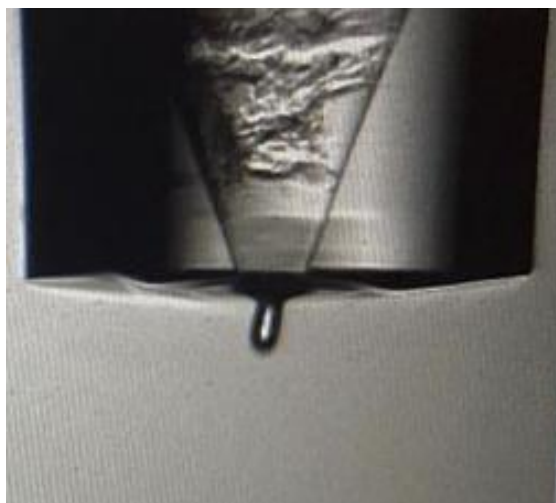


Figure 29. This resulting image shows a drop that cannot escape the orifice due to high surface tension when the dwell voltage was decreased from 54 V to 50 V while keeping the dwell time at 9 μ s

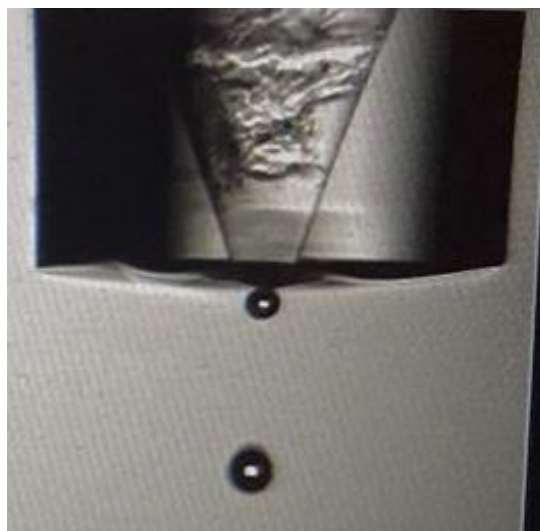


Figure 30. This resulting image shows a drop that is followed by a smaller drop that sometimes would pinch-off when the dwell voltage was increased from 54 V to 62 V while keeping the dwell time at 9 μ s

If the dwell voltage is below the optimal dwell voltage value, the toluene starts to form a droplet but does not overcome the surface tension forces to eject from the orifice and is sucked right back into the nozzle. But if the dwell voltage is greater than the optimal dwell voltage, the toluene dispenses a primary drop followed by a secondary smaller drop that is absorbed back into the nozzle as well. Awareness of these patterns may be beneficial for comparing results with future work.

Chapter 5: Conclusions and Future Work

The manual waveform tuning experiments demonstrated that toluene is jettable in piezo inkjets and yielded a drop volume of 30.2 pL. Once the inkjet system was subjected to a contaminant trapped within the nozzle of the inkjet, the manual tuning process was repeated and confirmed that toluene was still jettable and yielded a drop volume of 23.49 pL instead. Regardless of the state within the inkjet nozzle, whether it was completely clean or partially clogged, toluene droplets were consistently generated without any satellite deformities. Moving forward, instead of using manual waveform tuning to find optimal waveform parameters to produce toluene drops, an automated tuning process should produce droplet characteristics set by the user. If a contaminant is present, but the system is still jettable then automated waveform tuning should still be able to produce those same desired drops. The additional insights found in this project could serve as a benchmark or as a potential general pattern for jetting nonpolar solvents. Particularly, the observation of the toluene behavior at the nozzle orifice when key parameters, such as dwell time and dwell voltage, were intentionally manipulated. A better understanding for the jetting of nonpolar liquids could help reduce calibration time and reduce waste during an automated process optimization.

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